

COVID-19 Burden vs Resource Access in San Francisco Neighborhoods

Group 29 Contributions

Megan Upshaw: Gathered and organized data from all three San Francisco open-data sources.

Cleaned and merged datasets to neighborhood geography. Created map visualizations and figures. Organized and wrote the report (methods, results, summary, references) and integrated final outputs/code.

Austin Wong: Added analysis and visualizations for death vs neighborhood access. Conducted nonparametric Kendall's Tau and linear regression slope hypothesis tests to determine whether or not a linear relationship existed between the variables case rate vs neighborhood access and death rate vs neighborhood access. Wrote the abstract and introduction to the report, and provided additional details on the methods hypothesis testing section.

Prepared for:

Professor Nicolai Amann

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I. Abstract

Repository link: <https://github.com/mgupshaw/STA-141B-Final-Project>

The project's goal is to determine if there is a significant correlation between the number of vaccine and testing sites for COVID-19 and case rates, as well as death rates, for the city of San Francisco. We accomplished this goal by first finding three appropriate datasets, which included information of vaccine sites, testing sites, and case and deaths in neighborhoods in San Francisco. By cleaning and joining the datasets based on neighborhood locations, we were then able to conduct various summary statistics, visuals, and hypothesis tests for the correlation and slope. Our tests showed us that although the observed data correlates total access and case/death rates, there is no significant correlation or slope, suggesting the variables are largely uncorrelated. These results may be an important factor to consider when designing plans to contain future pandemics.

II. Introduction

COVID-19 is a highly infectious disease which can cause severe illness in some people. This makes it dangerous to the elderly, the immunocompromised, and people with other health conditions. Thus, analyzing data on COVID-19 is very important to protect the vulnerable populations, as well as the average, healthy person. This report analyzes whether or not the number of testing sites and vaccine sites affects the case rates and death rates of COVID-19. This can be helpful to government officials and epidemiologists to understand the relationship between access and case/death rates in order to create more effective plans at containing future epidemics, and especially new strains of COVID-19.

Our three datasets come from data.sfgov.org, which is a site for open source data collected by the San Francisco City and County Departments. In particular, we chose a dataset for COVID-19 cases and deaths by geography, a dataset for COVID-19 testing locations, and a dataset for vaccine access points in San Francisco. To start our analysis, we must clean and join the datasets according to neighborhood and geometry (location) data. The relevant variables used in this project include geolocation data, area type, case count, case rate, and deaths. After filtering the cases dataset, we were working with data from 41 neighborhoods.

We hypothesized that the more access points (vaccine sites + testing sites) there are, the lower the case and death rate would be. So our null hypothesis would be that there is no

correlation between access points and cases, and access points and deaths, and our alternative hypothesis is that there is a correlation, and therefore the linear regression would have a nonzero slope. We will conduct appropriate correlation and linear relationship hypothesis tests to assess our claims.

III. Methods

To answer our research questions, we loaded the three COVID-19-related datasets from the San Francisco Open Data portal into pandas, cleaned and filtered the data to focus on Analysis Neighborhoods, used GeoPandas to align neighborhood polygons with testing and vaccine site locations, and constructed a neighborhood-level summary table with new variables. We used visualizations to explore and present the results.

Data Sources

The project uses three main datasets from data.sfgov.org:

- 1. COVID-19 Cases and Deaths by Geography**

This dataset reports cumulative confirmed cases, a case rate measure, total deaths, and an estimated population for each geographic unit, including an “Analysis Neighborhood” geography. A geometric description of the boundaries of each neighborhood is also included.

- 2. COVID-19 Testing Locations**

This dataset lists COVID-19 testing sites. It includes each site’s name, address, and geographic coordinates that can be mapped to a point on the city.

- 3. COVID-19 Vaccine Resource Locations**

This dataset lists vaccination sites. It includes each site’s name, address, and geographic coordinates that can be mapped to a point on the city.

Combining these three datasets allows for the description of COVID-19 outcomes as well as the number of testing and vaccination locations inside the boundaries of each analysis neighborhood.

Data Cleaning to Focus on Analysis Neighborhoods

Because the cases dataset includes multiple geographic units (such as the entire city, ZIP codes, and census tracts), the first step was to define a consistent unit of analysis. For this

project, we focused on Analysis Neighborhoods, which are commonly used by the San Francisco Planning Department and provide a natural spatial scale for comparing COVID-19 outcomes. We removed all other geographic types which were “Citywide”, “Census Tract”, and “ZCTA (ZIP Code Tabulation Area)” in order to only work with the rows where the geography type was labeled as “Analysis Neighborhood”.

Within the filtered dataset, we renamed the neighborhood identifier column so that each row had a clear neighborhood label. We kept only the variables needed for our analysis. This included the columns “neighborhood”, “acs_population”, “count”, “rate”, “deaths”, and “multipolygon” which represent neighborhood names, estimated population, total case count, case rate per 100,000 residents, and total deaths. We also used the boundary geometry stored as a test multipolygon. This produced a neighborhood-level table where each row represented one Analysis Neighborhood in San Francisco.

We then examined the distribution of these variables using summary statistics and checked for missing values. A small number of neighborhoods had missing case counts and/or case rates, and some had missing death counts. In order to account for this, when an analysis required one of these variables, we restricted attention to neighborhoods with non-missing values.

For the testing locations and vaccine access dataset, there were minimal cleaning steps. We kept the columns describing the site and the columns describing location information. This included “Name”, “Address” “Latitude”, and “Longitude”. Any irrelevant or duplicate fields were dropped.

Geospatial Processing:

A key part of our project was aligning three different sources of geographic information. This included neighborhood shapes, testing locations, and vaccine locations. In order to do so, we used GeoPandas which extends pandas to work with geometric objects. GeoPandas implements GeoSeries and GeoDataFrame types which are subclasses of pandas.Series and pandas.DataFrame respectively.

In the case dataset, each Analysis Neighborhood had its boundary stored as a text multipolygon. We converted these text strings into polygon geometries and created a

GeoDataFrame with one polygon per neighborhood. This provided a spatial layer representing the Analysis Neighborhood boundaries for San Francisco.

In the testing dataset, each site's location was stored as a text point. In the vaccine dataset, locations were given by latitude and longitude coordinates, which we converted into point geometries for each site and built additional GeoDataFrames. One was with a point for every testing site and the other was with a point for every vaccine site.

We positioned the neighborhood polygons, testing-site points, and vaccine-site points in the same standard latitude/longitude reference system to ensure that all of these layers could be combined. The neighborhood to which each site belonged was then identified using a spatial join, more precisely a point-in-polygon procedure. For each testing and vaccination site, we located the Analysis Neighborhood polygon containing the site's point and added the neighborhood name to the site record. As a result, a neighborhood label that matched the neighborhood labels in the case data was included in both the testing and immunization datasets.

Merging

With neighborhood labels attached to each site, the next step was to construct access measures at the neighborhood level. The site-level data was aggregated to achieve this. A variable that keeps track of the number of testing locations in each neighborhood was created by grouping and counting all testing sites within the same area. The number of vaccination sites in each neighborhood was determined by repeating the same grouping and counting procedure for vaccine sites.

The neighborhood cases database was then updated with these counts. A count of zero was given to neighborhoods without any testing or vaccination locations. A single neighborhood-level dataset with one row per Analysis Neighborhood that included population, total cases, case rate, deaths, number of testing sites, and number of vaccine sites was produced as a result of the merging process.

New Variables

Two key derived measures were created from this merged dataset. In order to capture overall access, we constructed a combined access variable, denoted here as total access, defined as the sum of the number of testing sites and the number of vaccine sites in each neighborhood.

This variable treats testing and vaccination as complementary access points and provides a single summary measure of how many COVID-related sites are located in a neighborhood. Then, we created a mortality rate variable defined as the number of COVID-19 deaths per 100,000 residents in order to place deaths on a comparable scale across areas with varying population sizes. This was computed by dividing the total number of deaths in a neighborhood by its population and multiplying by 100,000. More insightful comparisons between neighborhoods of various sizes are possible when a rate is used instead of raw counts.

Hotspot Classification

In order to classify neighborhoods with high COVID-19 burden and low access, we created categorical “hot spot” labels based on the distribution of case rates and access. We used quartiles to define what would count as a high/low case rate and high/low access. High-case neighborhoods were defined as those with case rates that were at or above the 75th percentile of the case-rate distribution. Low-access neighborhoods were defined as those located at or below the 25th percentile of the total-access distribution. Using these two cutoffs, each neighborhood was grouped into one of several categories, such as “high cases and low access,” “high cases but not low access,” “low access but not high cases,” or “other.” This classification was then repeated using the constructed death-rate measure in place of case rate to identify neighborhoods with high deaths and low access. These hotspot labels were merged back into the geographic neighborhood table and later used to create maps that show where high-burden, low-access neighborhoods are located. Some helper code for generating the hotspot maps was adapted from code generated with AI, but all analysis steps and interpretations in this report are our own.

Exploratory Visualizations

Several visualizations were produced to summarize and explore patterns in the data: top-burden bar charts, access distribution plots, burden vs access scatterplots, and hotspot maps.

Top-Burden Bar Charts: Neighborhoods were sorted by case rate and plotted in descending order to identify those with the highest cumulative COVID-19 burden. A similar plot was created for death rates. These bar charts pictured in the results section clearly show which neighborhoods experienced the highest case rates.

Access Distribution Plots: The distribution of testing counts, vaccine counts, and total access across neighborhoods was examined to understand how evenly or unevenly access sites were distributed.

Burden vs Access Scatterplots: To explore the relationship between COVID-19 burden and access, scatterplots were constructed with case rate on the vertical axis and total access on the horizontal axis. An analogous scatterplot was created for death rate versus total access (*Figure 3* in the results section). A simple least-squares line was added to each scatterplot to visualize the direction and strength of any apparent linear trend.

Hotspot Maps: Using the hotspot labels described earlier, maps were created in which neighborhoods were colored according to their combination of burden and access. One map displayed case-based categories (*Figure 4*), and another displayed death-based categories (*Figure 5*). These maps show whether high-burden, low-access communities are more evenly distributed or concentrated in specific areas of the city.

Before any formal statistical tests are used, these visualizations together offer a descriptive overview of how COVID-19 outcomes and access are spread throughout San Francisco neighborhoods.

Statistical Analysis

After constructing the neighborhood-level dataset and exploring it visually, we used several statistical methods to quantify the relationship between COVID-19 burden and access.

We computed correlation coefficients between case rate and total access, and between death rate and total access, using the subset of neighborhoods with complete data. Correlation measures the strength and direction of linear association. Values close to zero indicate a weak linear relationship, while positive or negative values indicate a tendency for higher access to be associated with higher or lower outcomes respectively.

However, as later discussed in the results section, both the access variables and total access are skewed. The scatterplots also indicate that the relationships may not follow a simple straight line and may be influenced by outliers. To address this, we also used Kendall's tau, a nonparametric measure of monotone association (Pouokam). This is a good test for significant correlation when outliers exist and the data is skewed as it measures the number of concordant and discordant pairs of data for every combination of pairs. Then, the test permutes the data to

create a distribution which can then be used for hypothesis test decision making. Kendall's tau was computed for four pairs of variables: the number of testing sites and case rate, the number of testing sites and death rate, the number of vaccine sites and case rate, and the number of vaccine sites and death rate. For each pair, a one-sided hypothesis was specified based on whether we expected more sites to be concentrated in high-burden neighborhoods or associated with lower burden. The resulting tau statistics and p-values were used to assess whether there was evidence of a consistent monotone relationship between access and outcomes across neighborhoods.

To obtain a simple model of how COVID-19 burden changes with access, we fit simple linear regressions with case rate as the outcome and total access as the predictor, and with death rate as the outcome and total access as the predictor. These regressions estimate an average slope that describes how much the case or death rate changes, on average, when total access increases by one additional site. The associated p-values test whether the estimated slopes differ significantly from zero at conventional significance levels.

By combining correlations, Kendall's tau tests, and simple linear regressions with the descriptive plots and hotspot maps, we were able to provide multiple and complementary perspectives to answer the question whether neighborhoods with more COVID-19 testing and vaccine access points experienced lower COVID-19 case and death burdens.

IV. Results

The merged neighborhood dataset revealed significant differences in COVID-19 access and outcomes between individual neighborhoods. When neighborhoods are sorted by case rates, the ten highest case-rate neighborhoods are Bayview Hunters Point, Portola, Visitacion Valley, Mission Bay, Excelsior, Tenderloin, Mission, Outer Mission, Western Addition, and Potrero Hill as pictured in (*Figure 1*). Other neighborhoods have significantly lower case rates, making these neighborhoods stand out in relation to the citywide distribution. These communities are near the upper tail of the distribution, while many other neighborhoods have significantly lower case rates, as shown by the whole horizontal bar chart of case rates for all neighborhoods (*Figure 7*).

The constructed death rate per 100,000 residents also varies. Its distribution is somewhat smaller than the case rate distribution though as pictured in (*Figure 8*). Some of the neighborhoods that rank high in *Figure 7* also appear near the top in *Figure 8*. However, the rankings are not identical. This suggests that while the case rates and death rates are related,

additional neighborhood characters are likely also influential and are not fully captured by our access measures.

The relationship between COVID-19 and total access is visualized through scatterplots of case rates and death rates versus total access. Instead of a clear trend, the scatterplot of case rate against total access (*Figure 2*) displays scattered points. Rather than being concentrated in the lowest-access group, neighborhoods with the greatest case rates are dispersed over a variety of access levels. The fitted line in *Figure 3* has a modest positive slope of 36.503865160224166 when a least-squares line is added, but the points are widely dispersed around it. This suggests that the number of testing and vaccination locations in an area is relatively weakly correlated with case rates. An even smaller slope of -4.095706565751768 is seen in the death-rate scatterplot with best-fit line (*Figure 4*). The fitted line is weakly negative, and neighborhoods with lots of access points occur throughout a wide range of fatality rates. This suggests that total access explains very little of the variation in deaths across neighborhoods.

The hotspot maps provide a complementary, categorical perspective on how burden and access align. In the case-based hotspot map (*Figure 5*), neighborhoods are classified according to whether they fall into high or non-high case-rate quantiles and low or non-low access quantiles. Only a small number of neighborhoods fall into the “high cases, low access” category, while several of the highest case-rate neighborhoods appear in groups with moderate or high access. The death-based hotspot map (*Figure 6*), which applies the same logic using death rates instead of case rates, shows a similar pattern. A few neighborhoods combine relatively high death rates with low access, but many high-death neighborhoods have moderate or high access counts. Together, *Figure 5* and *Figure 6* suggest that although some neighborhoods plausibly warrant concern as “high burden, low access,” there is no simple citywide pattern in which all high-burden neighborhoods are systematically underserved.

These visual findings are consistent with the numerical summaries. Correlation coefficients between total access and case rates and correlation coefficients between total access and death rates were not statistically significant at the 5% level. Nonparametric Kendall’s tau tests, which assess monotone association, indicate no significant relationship between testing counts and case rates ($p = 0.11388780712624114$), testing counts and death rates ($p = 0.7718488564937017$), vaccine counts and case rates ($p = 0.5$), and vaccine counts and death rates ($p = 0.364420335431793$). Simple linear regressions of case rate on total access (slope $p =$

0.732) and death rate on total access (slope $p = 0.644$) produce slopes that are small and not statistically different from zero. Overall, *Figures 1–8* and the associated test statistics all point to the same conclusion: across San Francisco’s Analysis Neighborhoods, neighborhood-level counts of COVID-19 testing and vaccine sites have limited power to explain which areas experienced the highest COVID-19 case and death burdens.

V. Summary

This project investigated if the death burden was lower in San Francisco communities with more access to COVID-19 testing and vaccination facilities. We created testing count, vaccine count, and total access variables, combined neighborhood outcomes with site locations, mapped burden and access, compared quarries, and assessed relationships. While relationships with mortality are almost nonexistent, estimated correlations between total access and case rates are weak but directionally positive. The outcomes hold true for all specifications. Two-sided tests were unable to reject the null hypothesis at $\alpha = 0.05$ in all cases. The results of this study show that cross-neighborhood variance in the COVID-19 burden cannot be adequately explained by testing and immunization site counts alone.

A couple of limitations should be taken into account when interpreting these findings. Among these are the following: effects accumulate over time while counts are a snapshot; counts are a generic measure that does not include capacity, hours, distance, uptake, etc.; and other potential cofounders, including population density, are left out. The evidence shows no correlation for mortality and a weak, statistically insignificant positive link for case rates within these bounds. To identify significant benefits, more sophisticated metrics of effective access are needed.

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List of Figures

Figure 1

Top 10 SF Neighborhoods by COVID-19 Case Rate

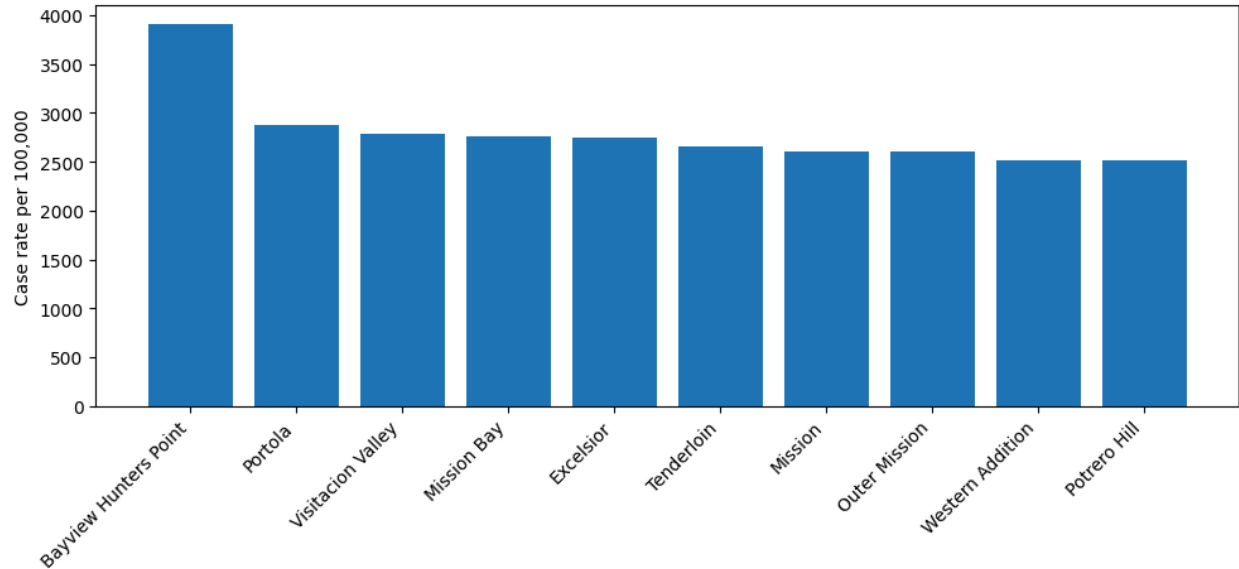


Figure 2

Case Rate vs Access Points by Neighborhood

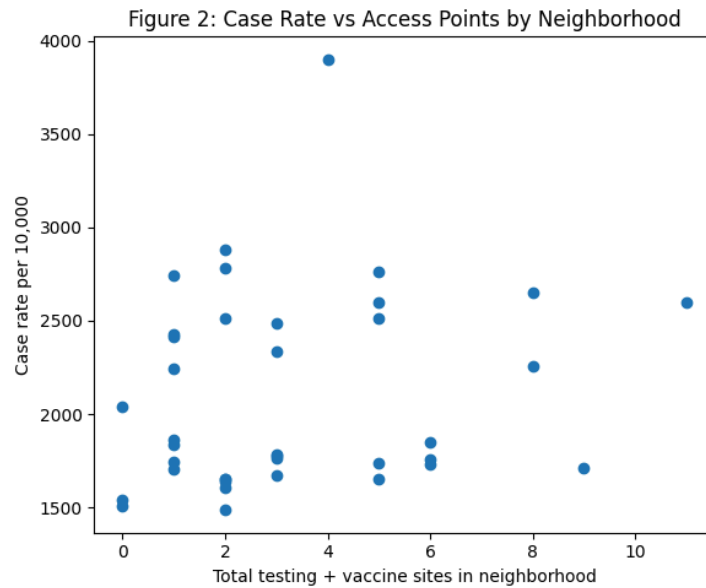


Figure 3

Case Rate vs Access with Best-Fit Line

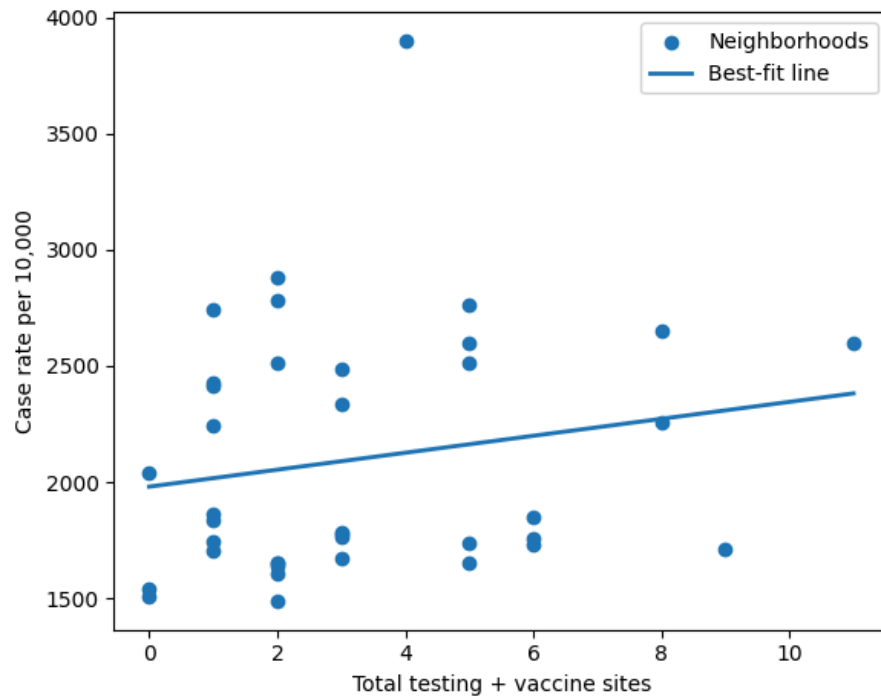


Figure 4

Death Rate vs Access with Best-Fit Line

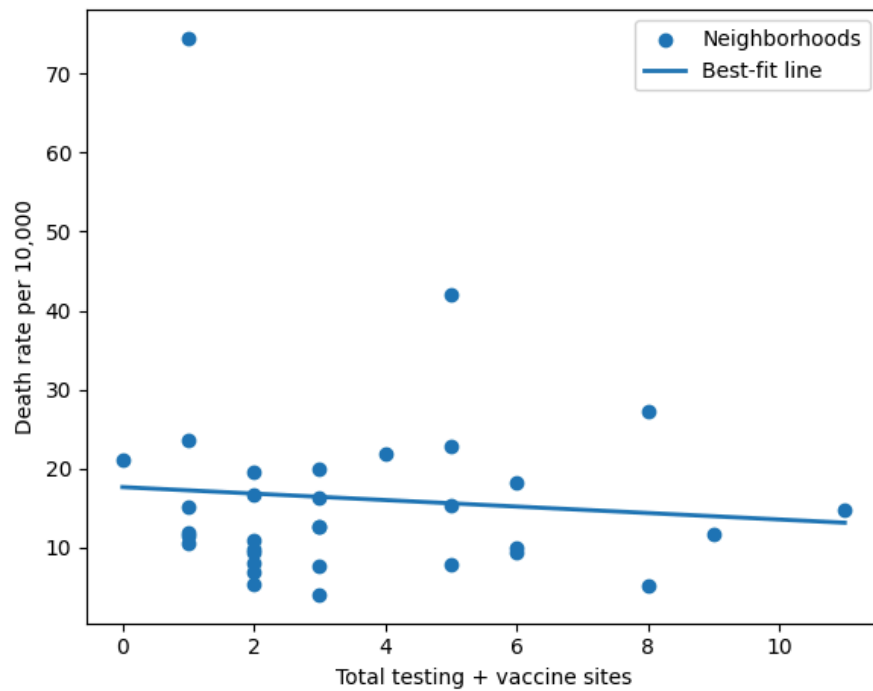


Figure 5

COVID-19 Hotspots: Case Rates and Access by Neighborhood

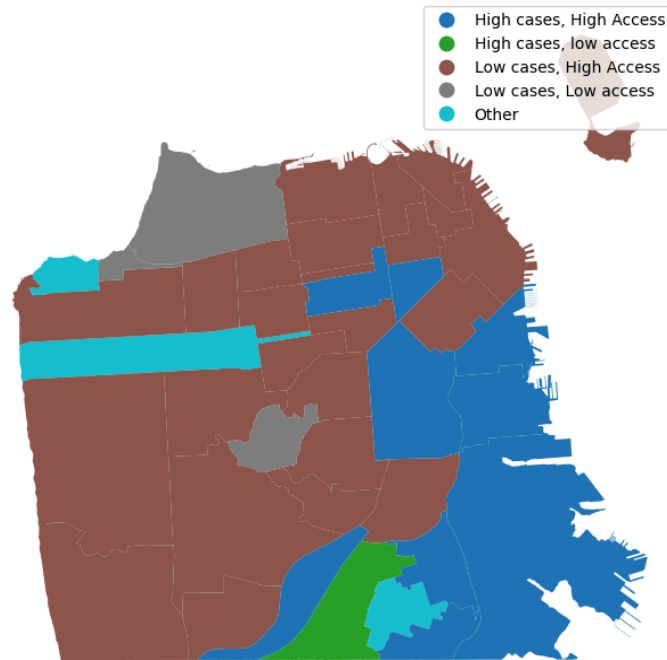


Figure 6

COVID-19 Hotspots: Death Rates and Access by Neighborhood

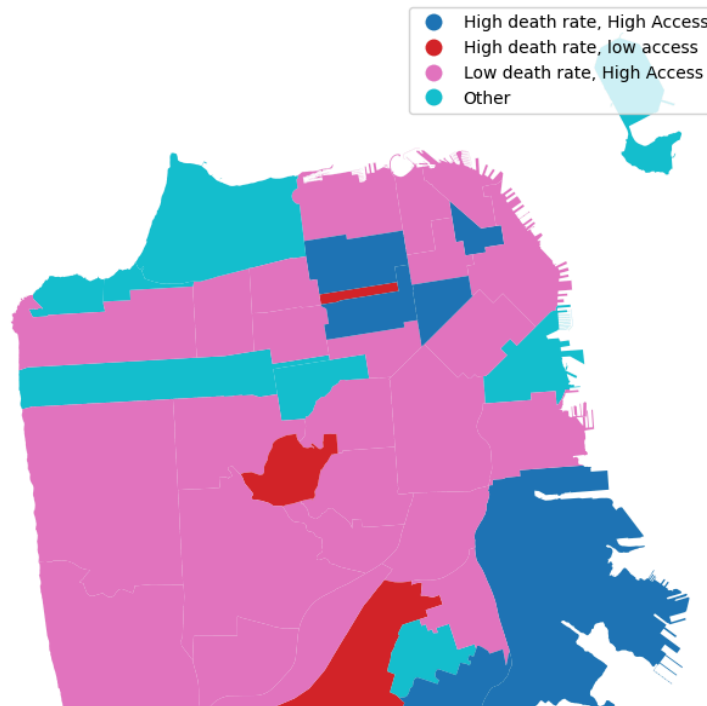


Figure 7
COVID-19 Case Rates by Neighborhood

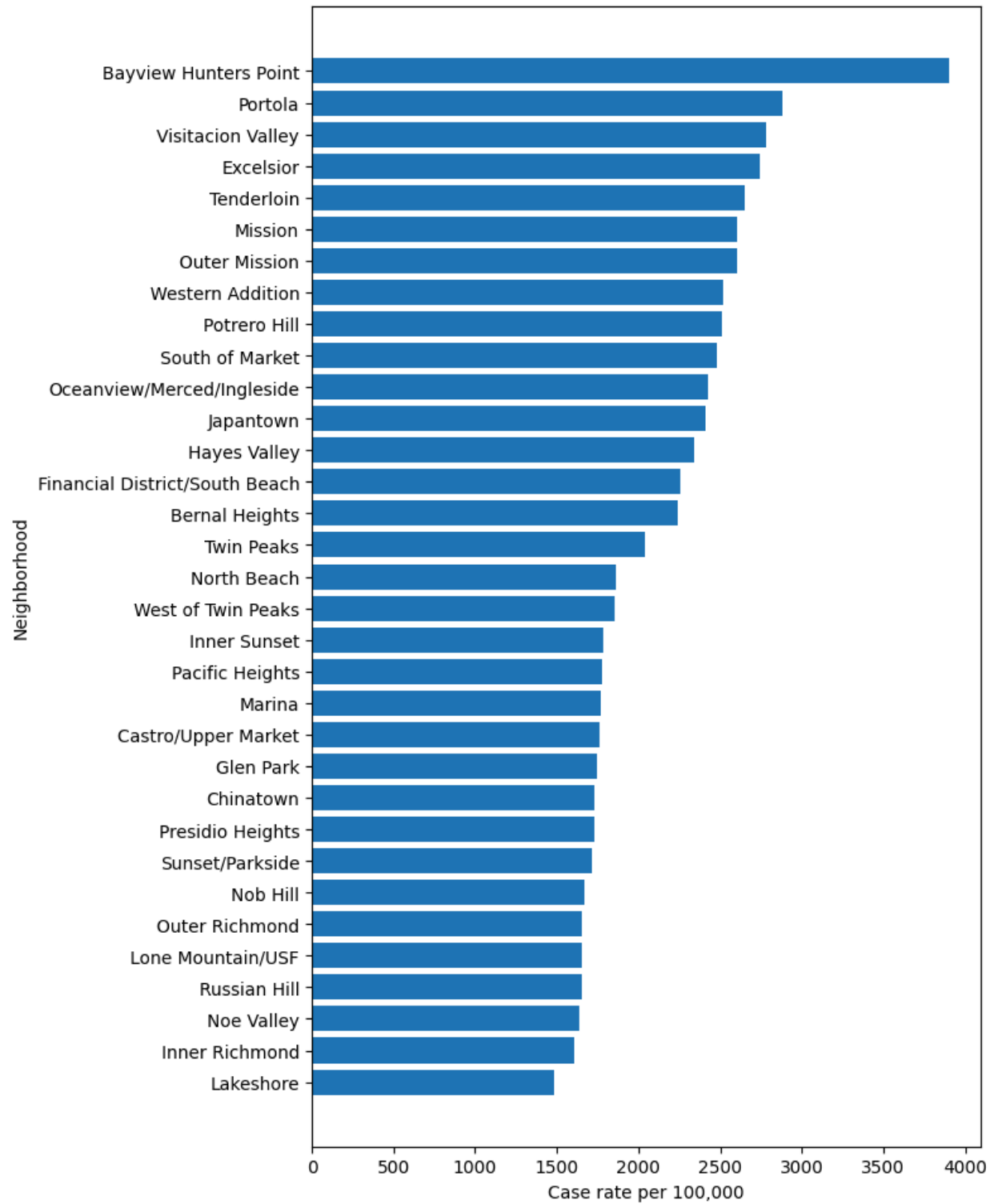
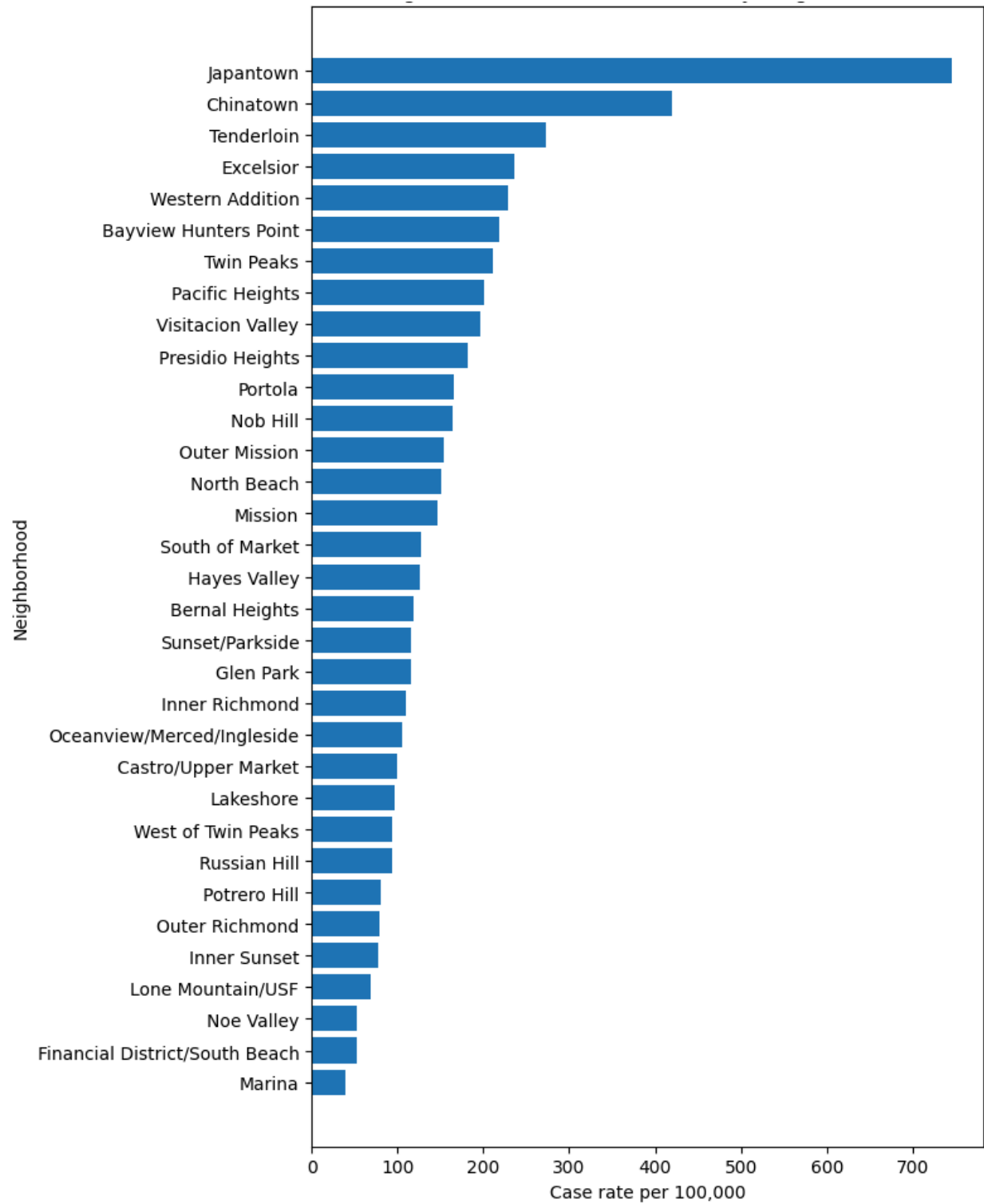


Figure 8

COVID-19 Death Rates by Neighborhood



Codes

Repository link: <https://github.com/mgupshaw/STA-141B-Final-Project>